MATLAB EXPO ONLINE

From Control Theory to Practice: Addressing Misconceptions in Control Design

Siddharth Jawahar, MathWorks



MathWorks®

Steps to follow along

- Look for the chat message with the MATLAB trial license and GitHub repo
- Click on the MATLAB trial license link and login into your MathWorks account (create one if you do not have one)



- Click on the GitHub link for the workshop files
- Navigate to the “Getting Started” section and click on 

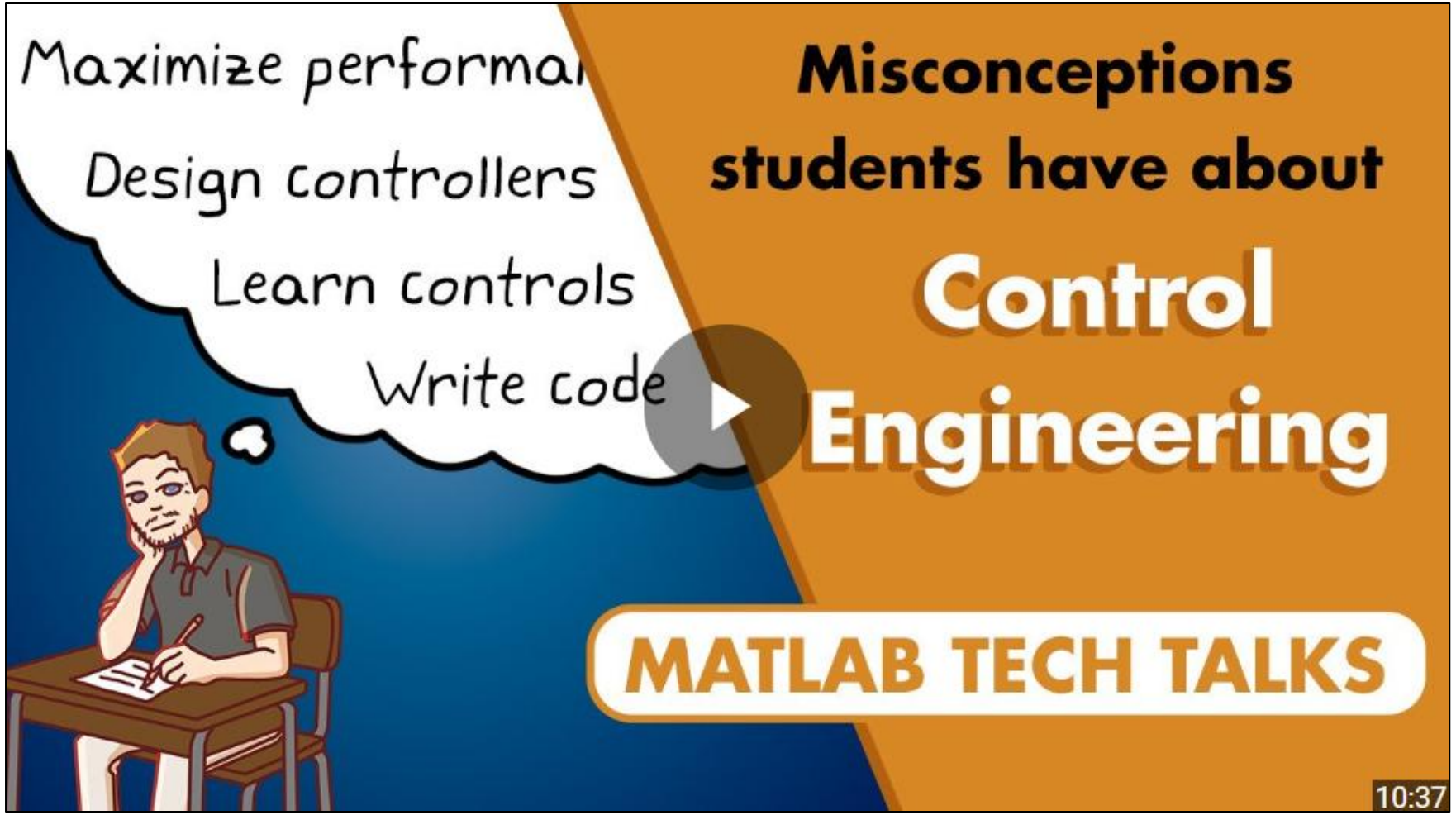
Agenda

- Introduction
- 5 misconceptions about controls engineering
  - I’m going to spend the majority of my time designing controllers
  - I need to maximize performance
  - I’m a control engineer so I only need to learn controls
  - I’m going to develop the design from scratch
  - The tools I’m using don’t matter
- CTA and Resources

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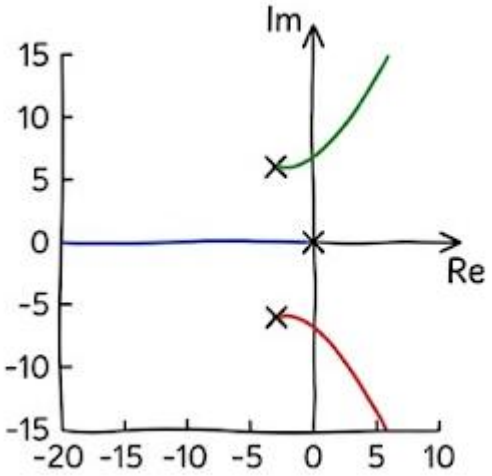
Brian Douglas' tech talk on misconceptions in control engineering



When you think of controls engineering, what pops into your head?



You WILL use these in industry, but it's just a smaller piece of the job

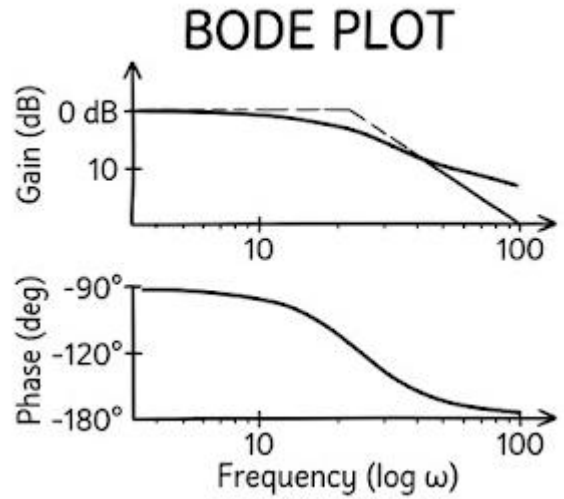


ROOT LOCUS

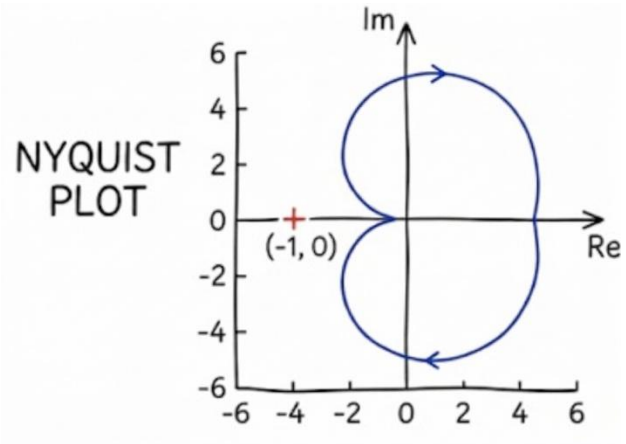
STATE SPACE

$$\dot{x} = Ax + Bu$$

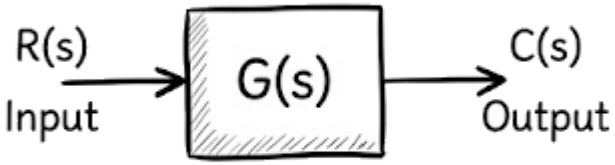
$$y = Cx + Du$$







TRANSFER FUNCTION



$$G(s) = \frac{Num(s)}{Den(s)}$$



r/AskEngineers • 4y ago

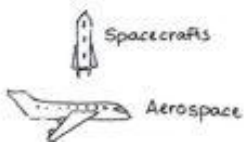
What does a control systems engineer actually do in said profession?

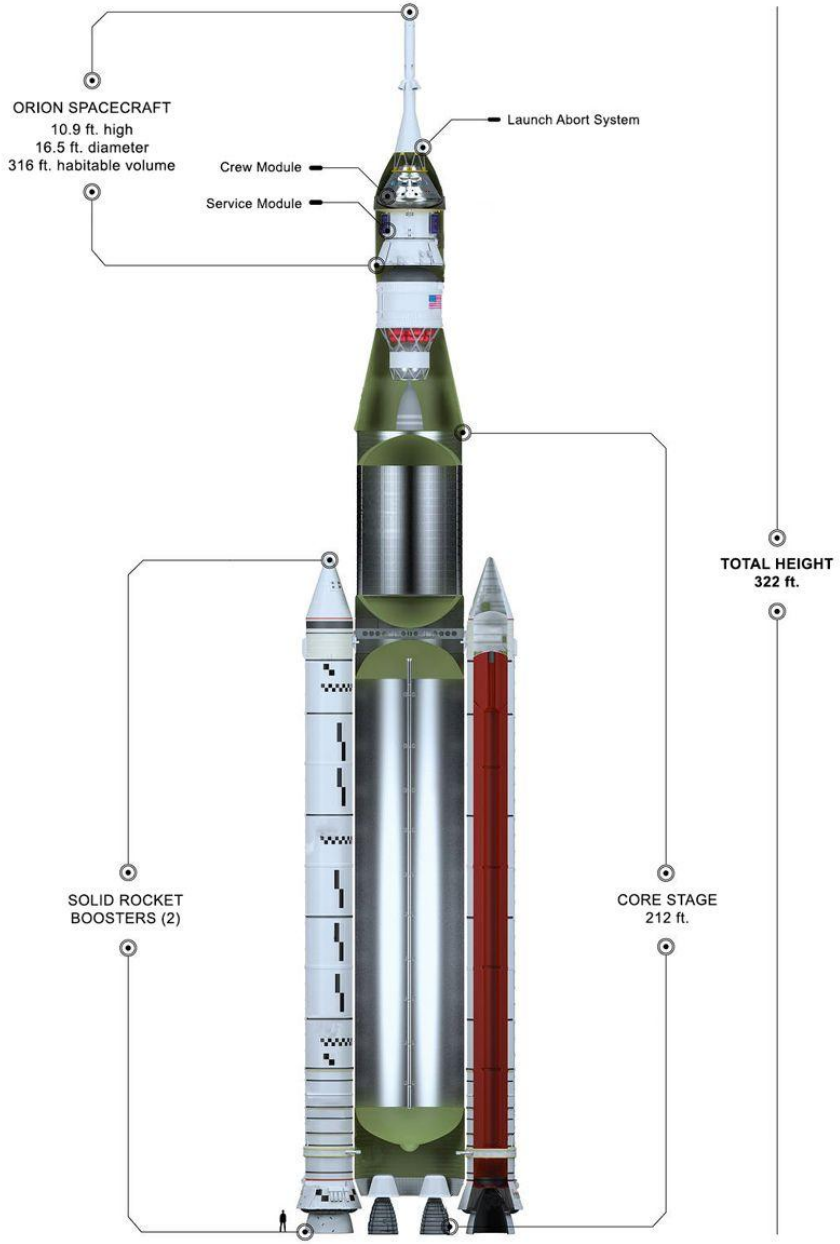
Electrical

I mean I have studied a bunch of theory stuff like stability, Nyquist plot, gain/phase margin, state transition matrix, controllability/observability, non linear systems, blah blah. They never taught us what do we actually do with that on real life.

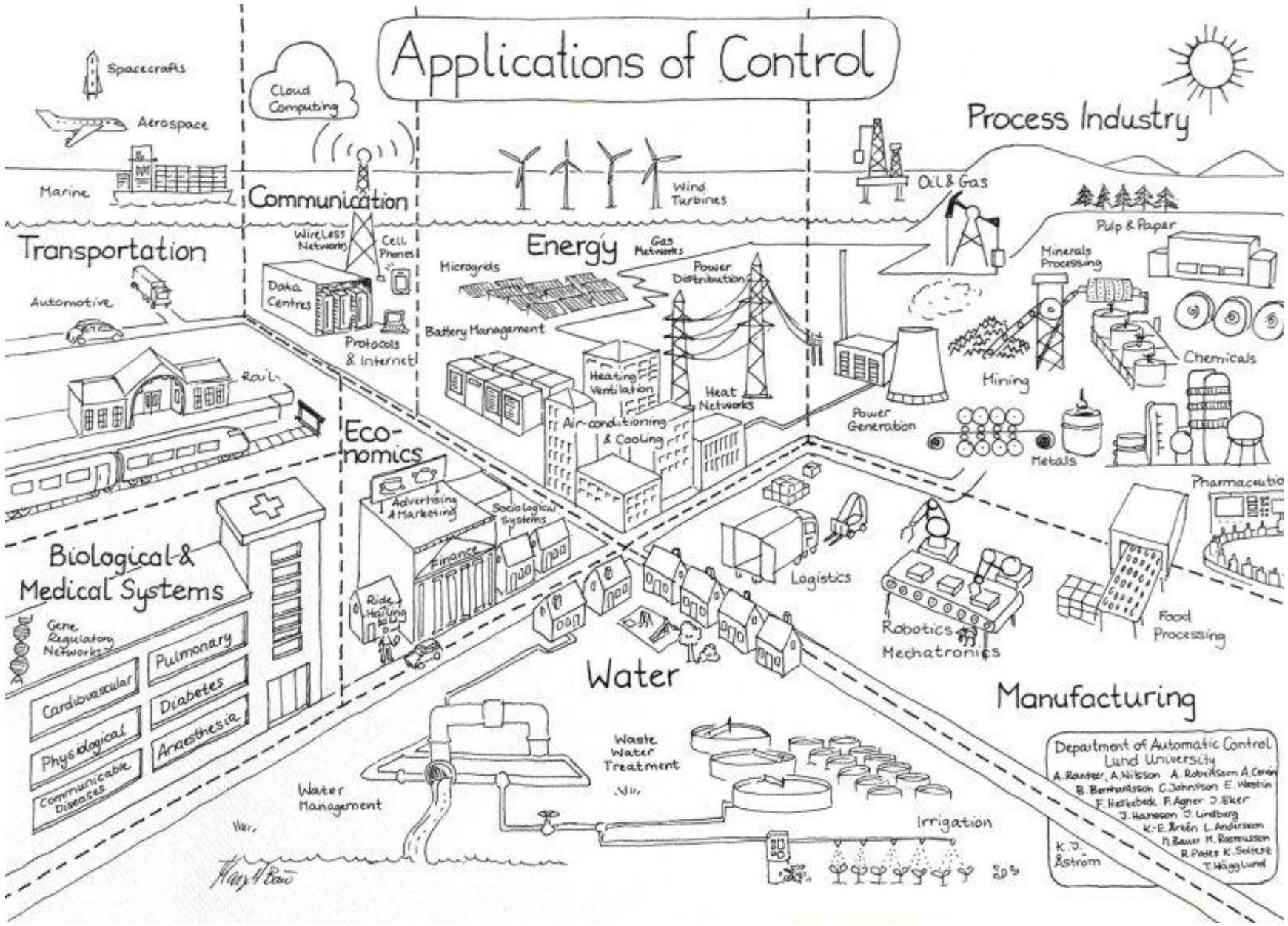
But what does a control systems engineer do with that in an actual job? I'm asking this because I have to get a masters, and I don't know if I should go for. I hate power systems, I love analog/digital electronics, neutral about electrical machine design, and I like the mathematical aspect of control systems and I understand that it is a huge multidisciplinary field.

EDIT : I know control systems engineer design/maintain "control systems", I wanna know what is the actual work that you do, what's a day in your engineering work life is like.

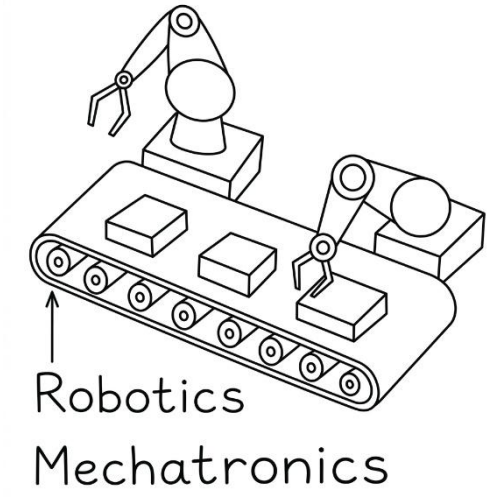




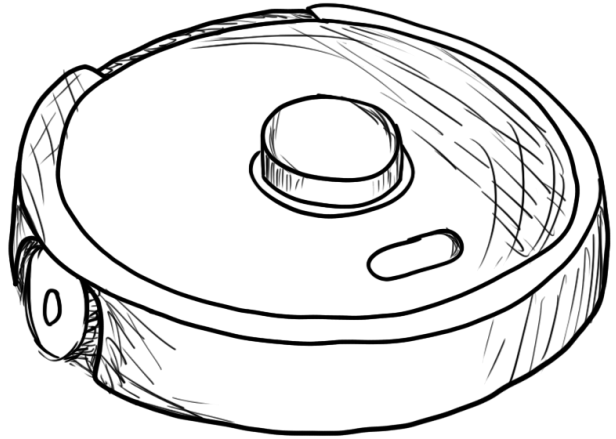
Artemis Rocket



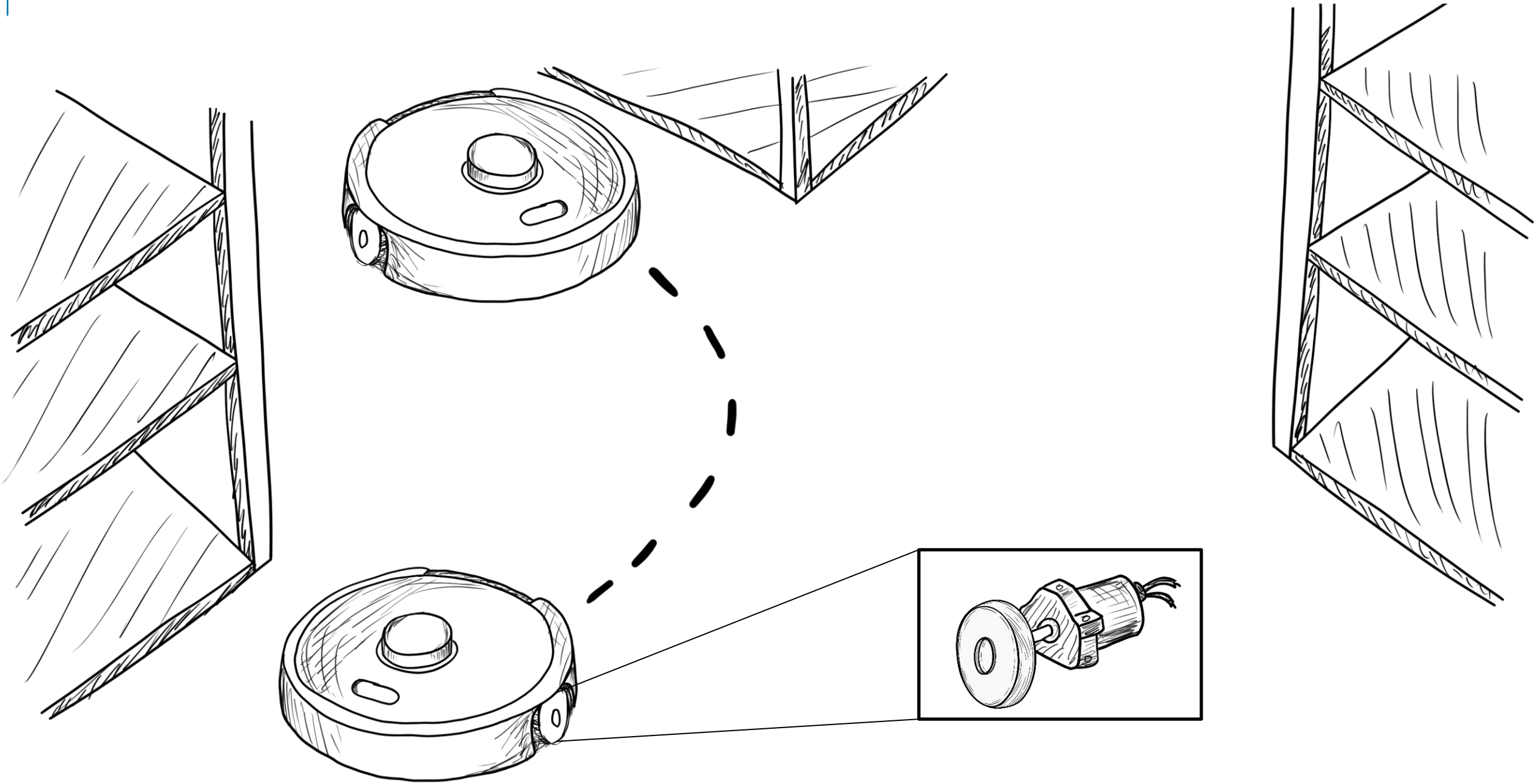
Drawing by Margret Bauer, Professor of Automation, HAW Hamburg



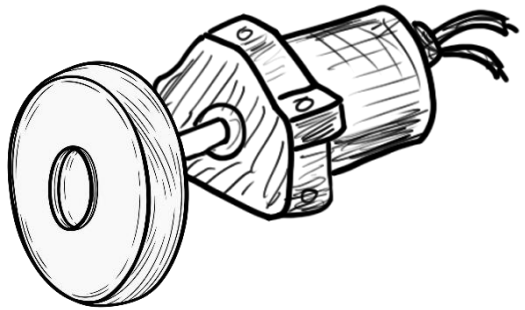
Differential drive robot



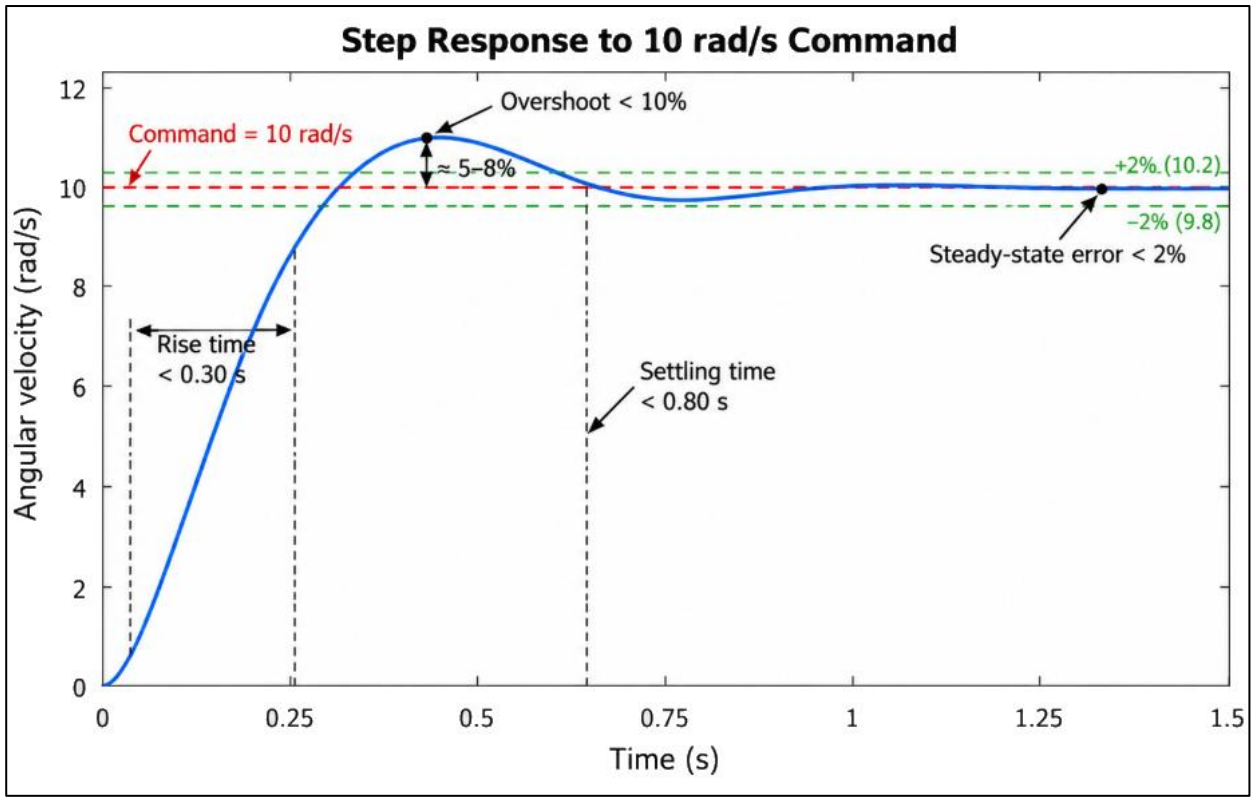
Applications – floor vacuums, lawnmowers, warehouses…



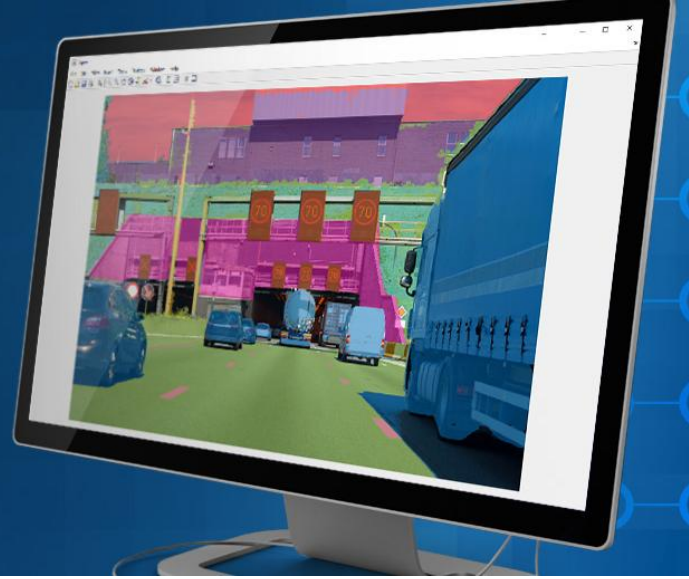
Task: Design a velocity controller for the DC motor



TIME DOMAIN REQUIREMENTS TO A STEP INPUT	
Step command	0 to 10 rad/s
Rise time	< 0.30 s
Settling time	< 0.80 s
Overshoot	< 10 %
Steady-state error	< 2 %
Motor voltage	within +/-12 V

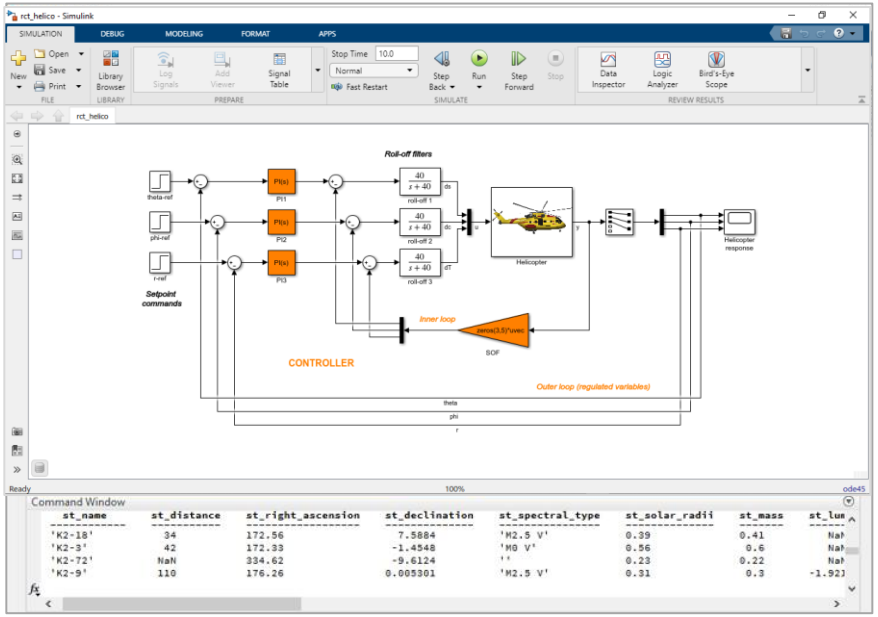


MATLAB®
&SIMULINK®



Our Products

- MATLAB is a programming environment for algorithm development, data analysis, visualization, and numeric computation.
- Simulink is a graphical environment for designing, simulating, and testing systems.



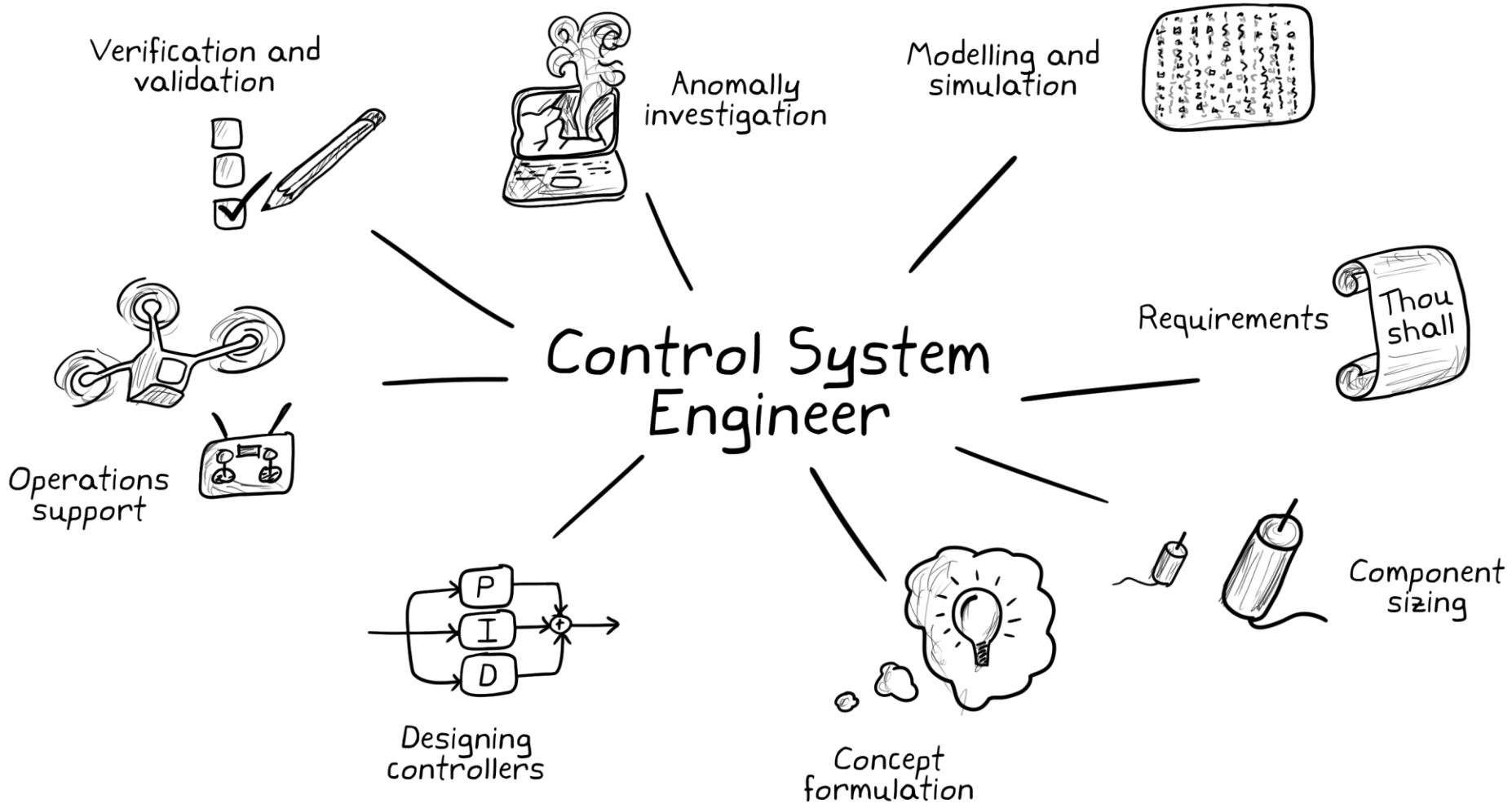
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Misconceptions I wish my controls class addressed

- I’m going to spend majority of my time designing controllers

Designing controllers is only a small part of the overall job

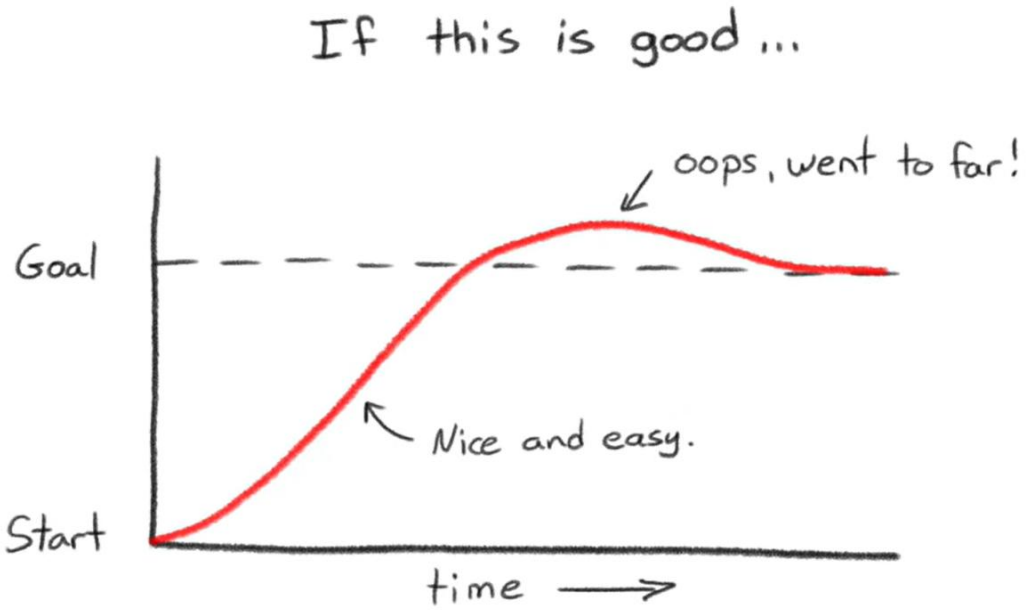


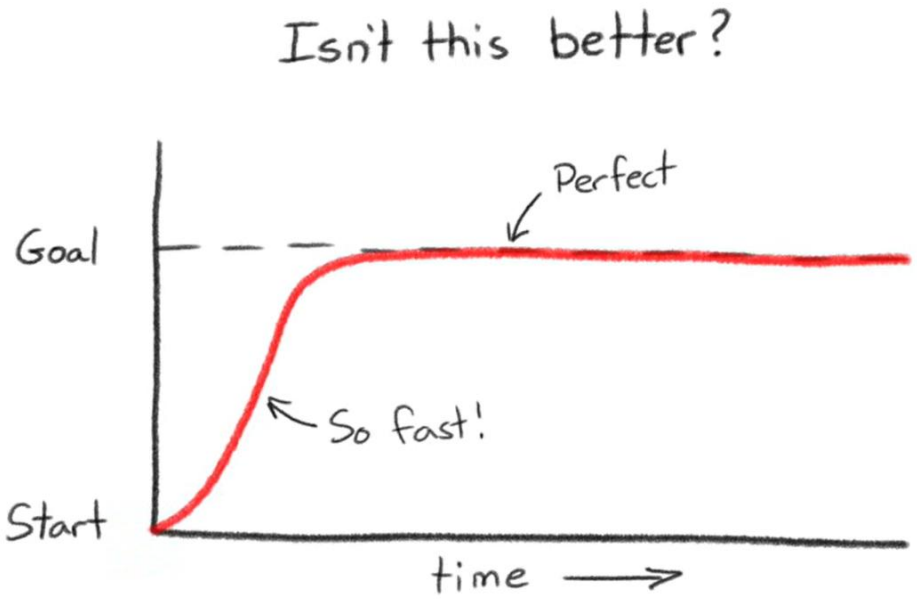
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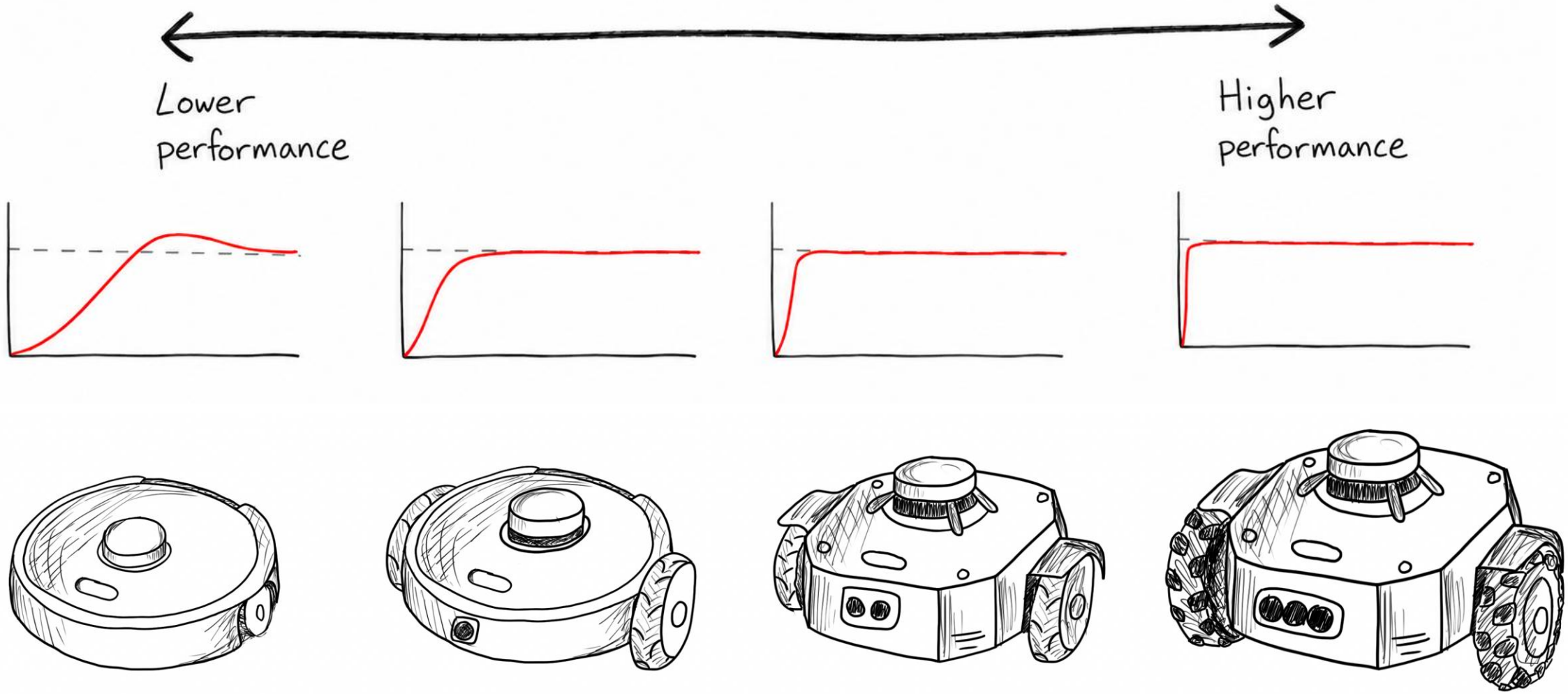
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- I need to maximize performance


If this is good ...
oops, went to far!
Goal
Nice and easy.
Start
time


Isn't this better?
Perfect
Goal
So fast!
Start
time



Increasing need for time & better controller hardware (sensors, actuators, microcontrollers)

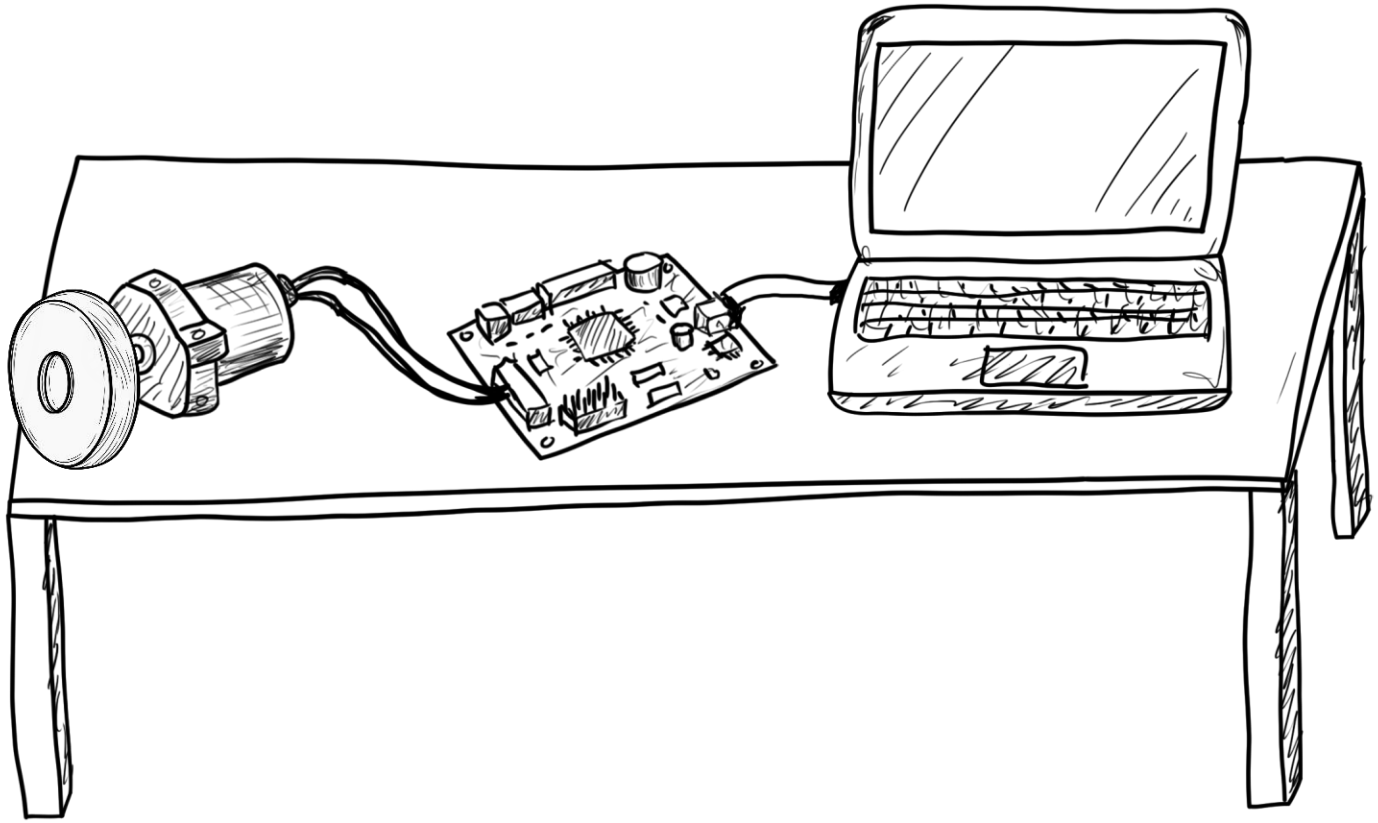
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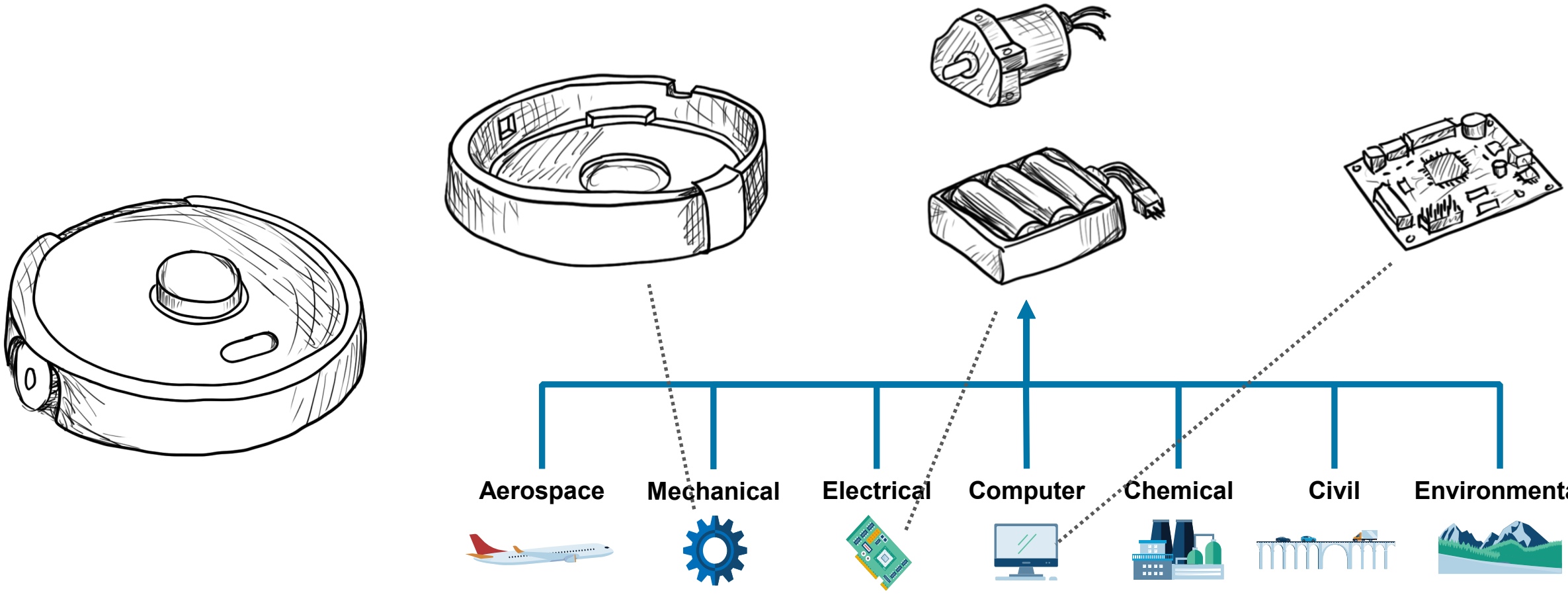
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Let’s test the controller on the motor hardware



Controls engineering requires system-level, multi-domain integration



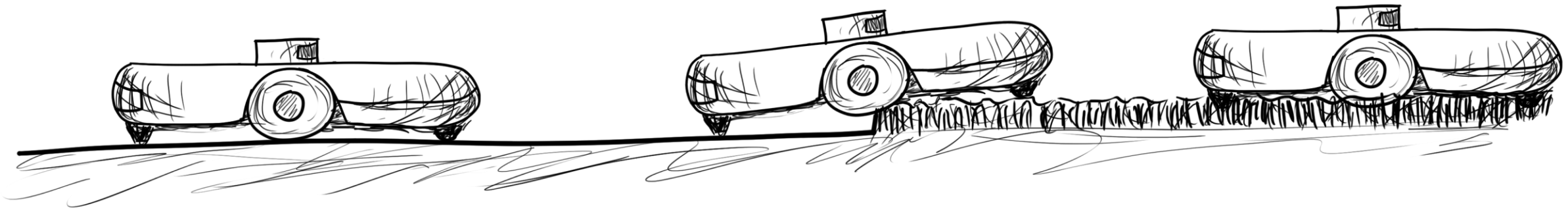
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Testing the robot on different floor surfaces



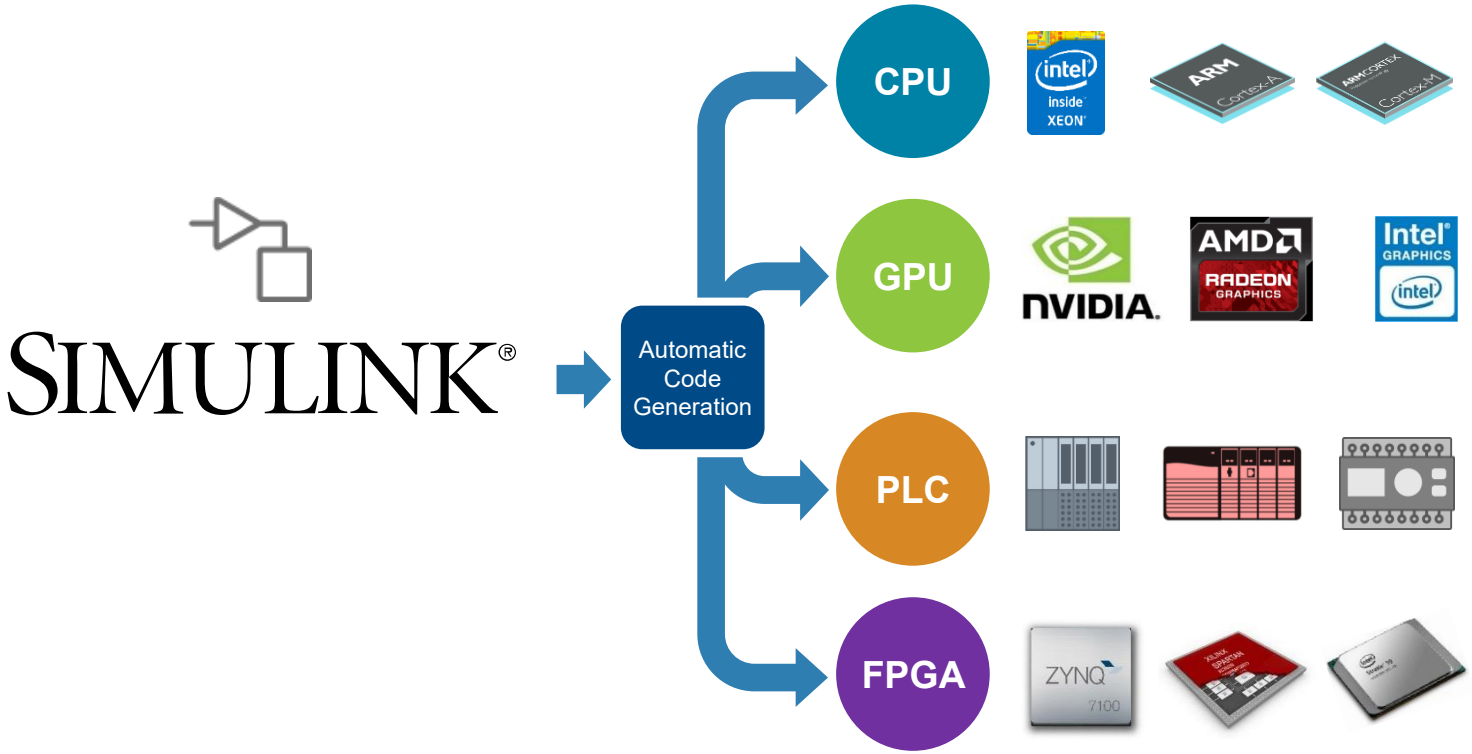
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Automatically generate code for prototyping and production



Reliable and **high performance**, with **flexible** choice of targets

German Aerospace Center (DLR) Robotics and Mechatronics Center Develops Autonomous Humanoid Robot with Model-Based Design

Challenge

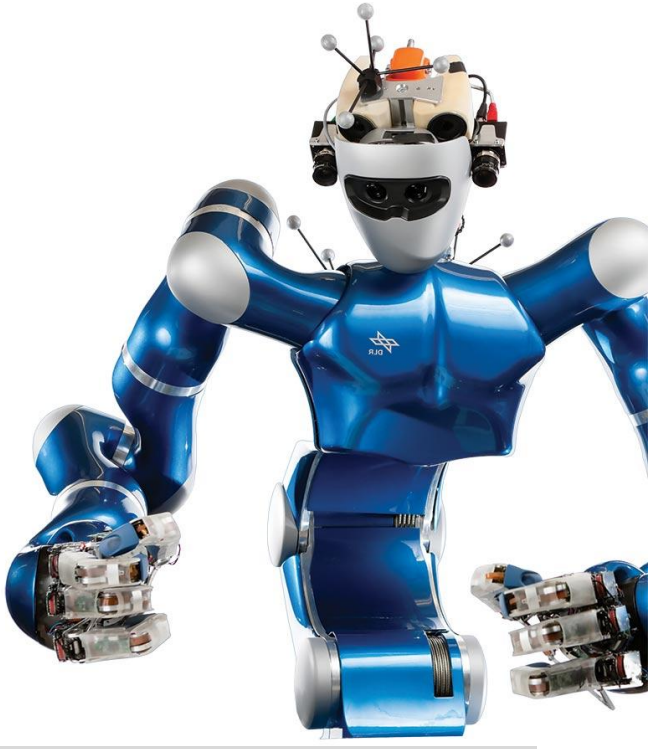
Develop control systems for a two-armed mobile humanoid robot with 53 degrees of freedom

Solution

Use Model-Based Design with MATLAB and Simulink to model the controllers and plant, generate code for HIL testing and real-time operation, optimize trajectories, and automate sensor calibration

Results

- Programming defects eliminated
- Complex functionality implemented in hours
- Advanced control development by students enabled



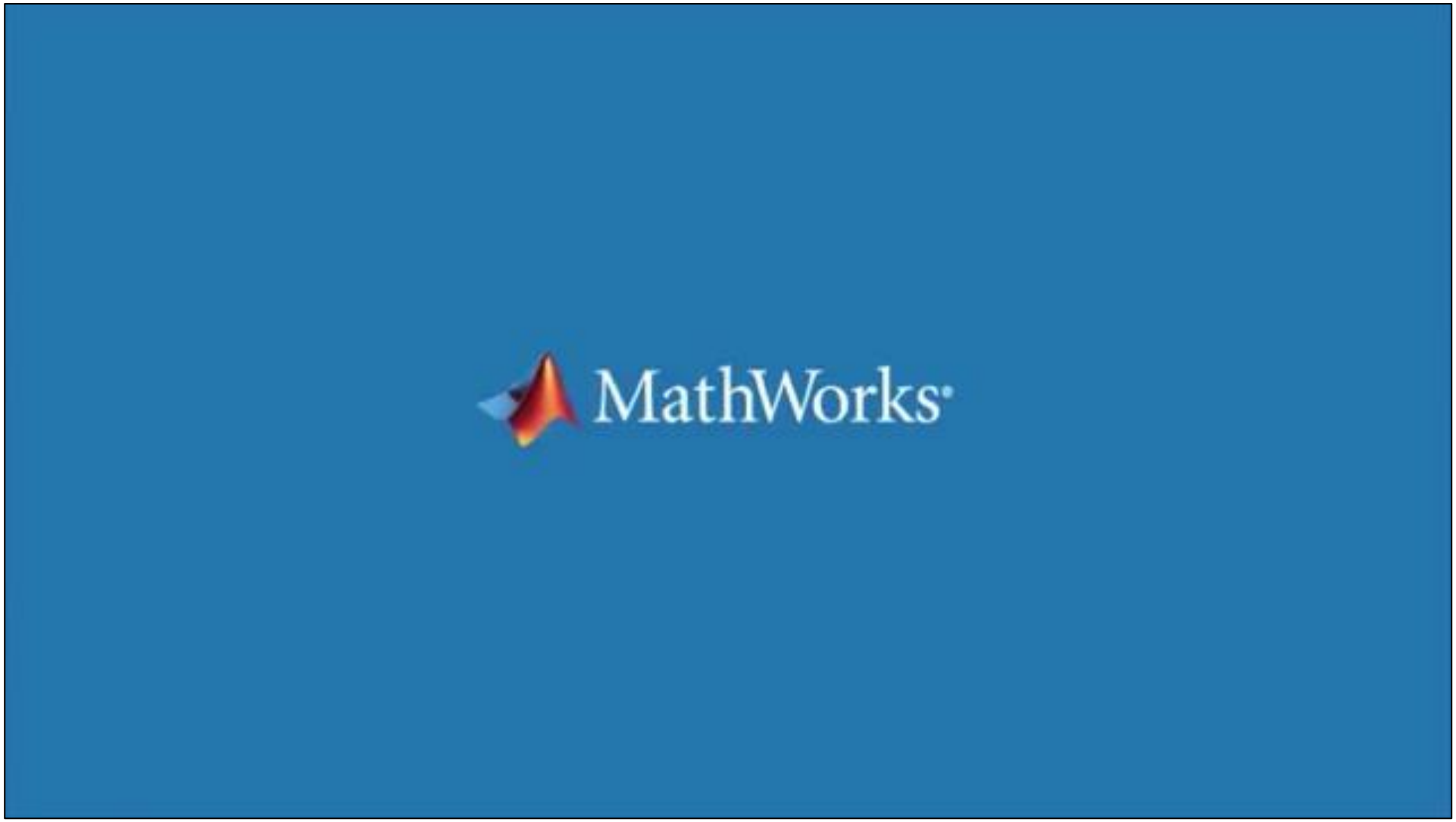
DLR's humanoid robot Agile Justin.

"Model-Based Design and automatic code generation enable us to cope with the complexity of Agile Justin's 53 degrees of freedom. Without Model-Based Design it would have been impossible to build the controllers for such a complex robotic system with hard real-time performance."

- Berthold Bäuml, DLR

Link to user story

German Aerospace Center (DLR) Robotics and Mechatronics Center Develops Autonomous Humanoid Robot with Model-Based Design



Accelerate your productivity with GenAI and MathWorks Tools

AI Assistants



MATLAB, Simulink and Polyspace Copilots

optimized for MathWorks Tools

Agentic Workflows



External GenAI Agents

use MathWorks tools with external Agents via MCP or VS Code



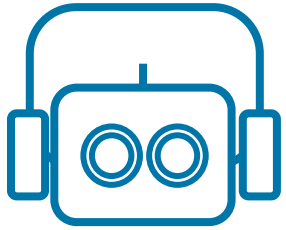
Build Agents in MATLAB

call LLMs from MATLAB code

Accelerate your productivity with GenAI and MathWorks Tools



Generative AI



Agentic AI

Simulink Copilot
optimized for Simulink

Simulink Agentic Toolkit
use Simulink with external AI Agents

R2026a

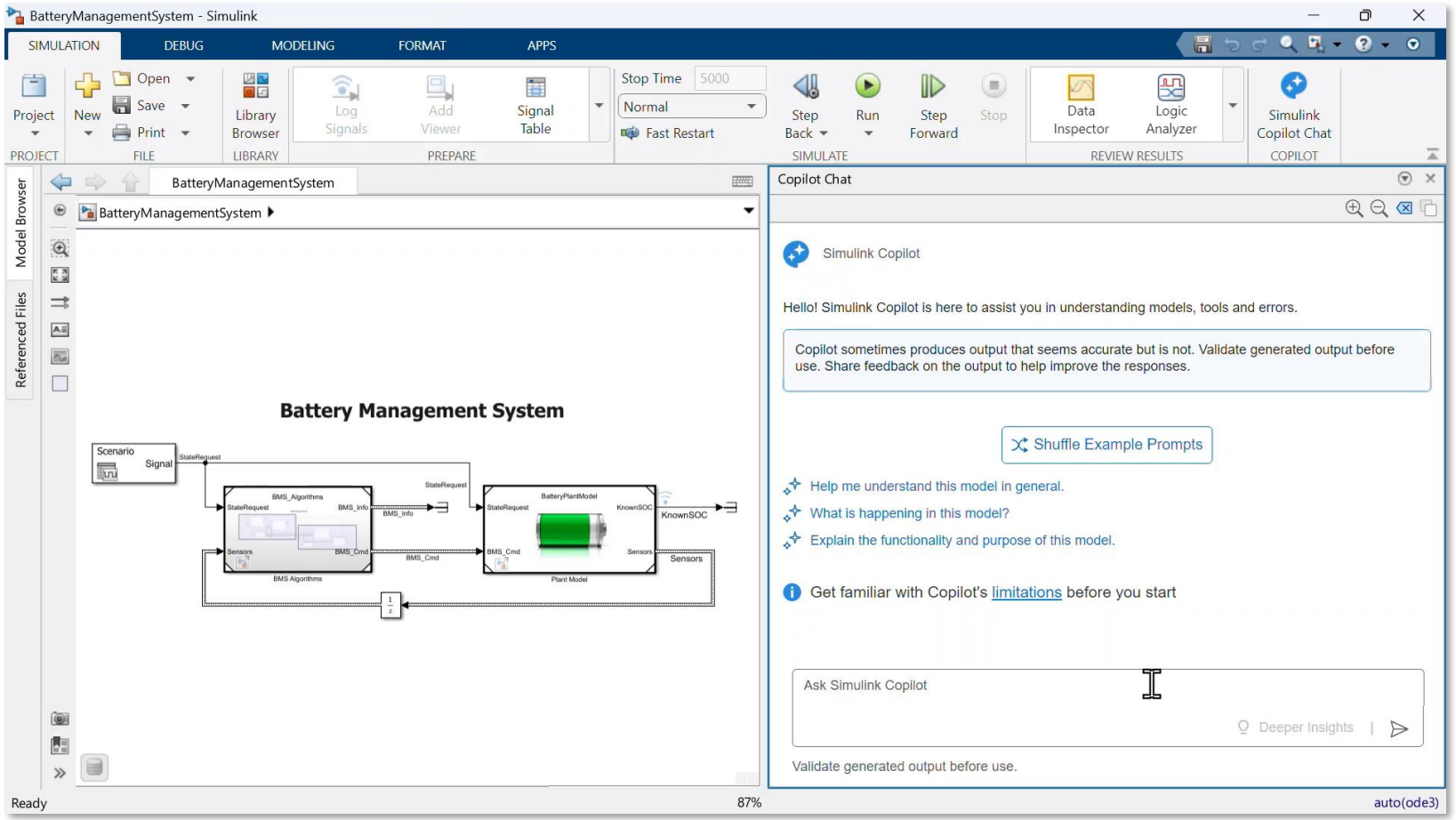
Simulink Copilot
Provides deep insights and actionable advice for Simulink models

Explain & Advise

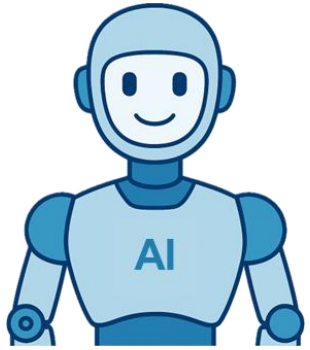
- **Learn tools & techniques**
- **Understand the model**
- **Search the model**
- **Get advice**
- **Troubleshoot errors**

Review & Automate

- **Automate tasks**
- **Track progress**



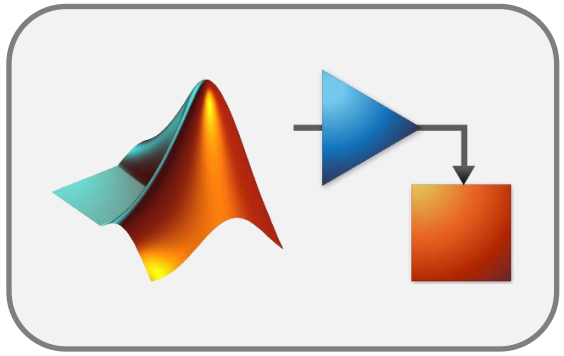
Agentic AI workflows with MathWorks tools are enabled by Model Context Protocol (MCP)



MCP Client

MCP

MCP Server



Cloud providers
OR
Local models

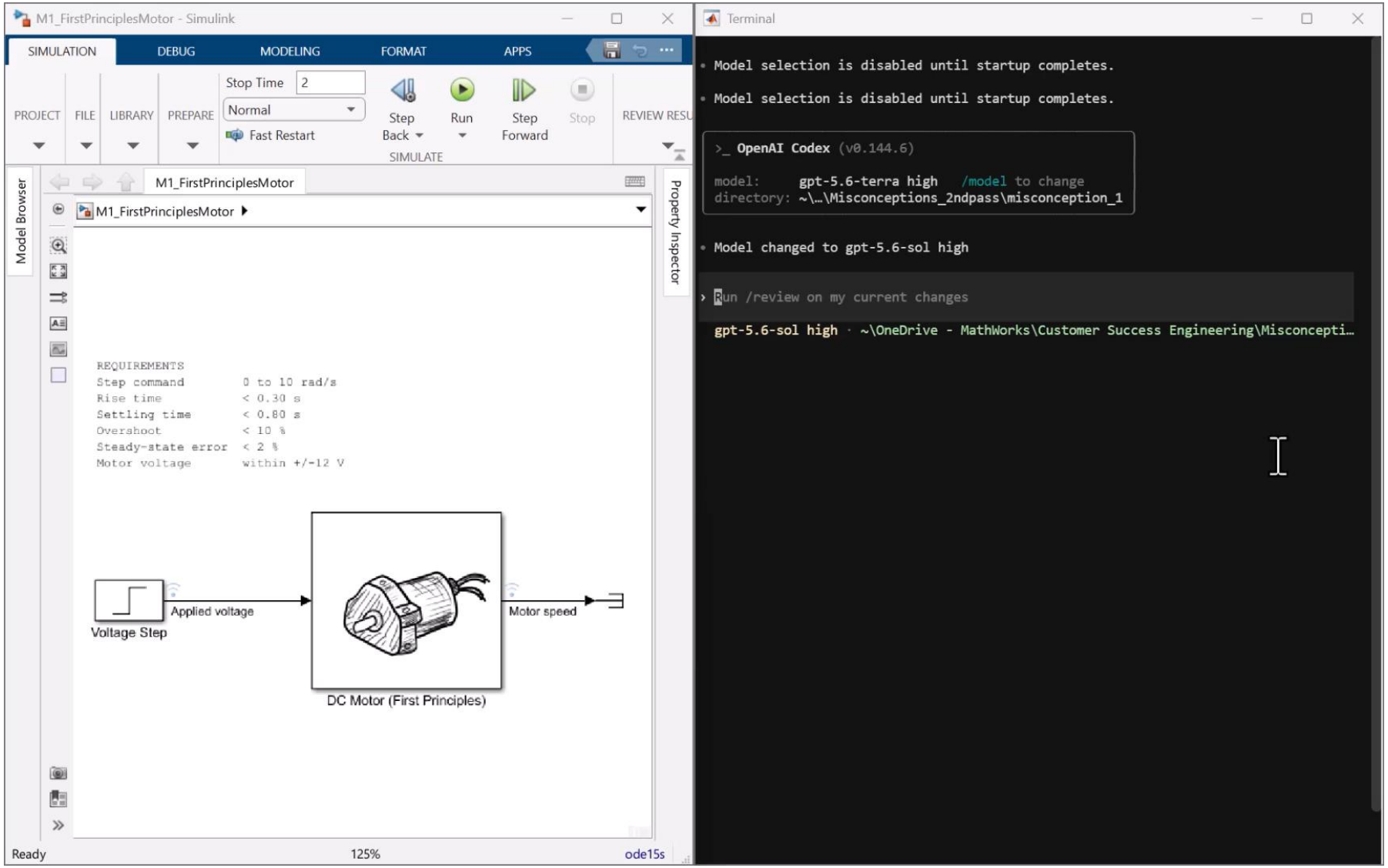
Gemini CLI

Open AI Codex

Claude Code

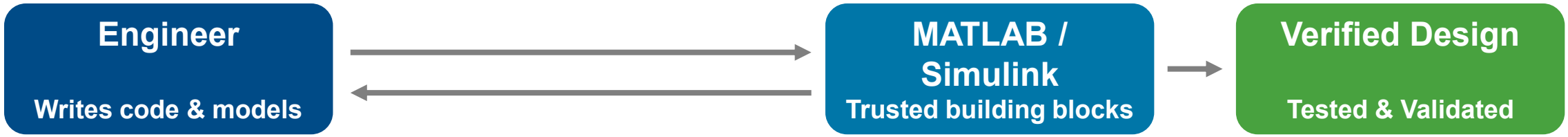
MATLAB & Simulink

Agentic AI workflow for tuning controllers

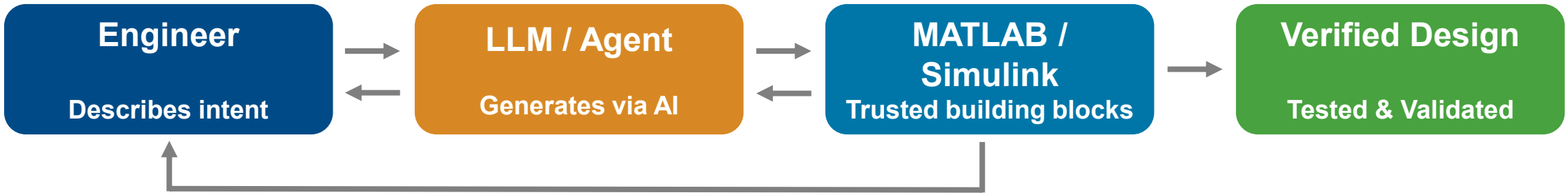


AI changes creation speed, not engineering accountability

Today



With AI



What doesn't change when you add AI

The engineer owns the design. The same engineering evidence verifies it.

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- I'm going to develop the design ~~from scratch~~ **efficiently**
- The tools I'm using ~~don't matter~~ **are important**

Summary - 5 misconceptions about controls engineering

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  - The tools I’m using don’t matter
- **CTA and Resources**

Call to action as you prepare for a career in controls engineering

- Strengthen your **engineering foundations** and **systems knowledge**
- Learn the tools that are important in industry
  - **AI literacy** is now a fundamental skill
- Get your hands dirty by **working with hardware (project-based learning)**

MathWorks supports deployment to a variety of low-cost hardware



Arduino



Lego EV3



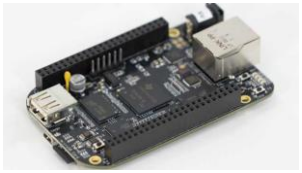
Raspberry Pi



Android/iOS Devices



Kinect for Windows



BeagleBone Black



Texas Instruments



STM Electronics



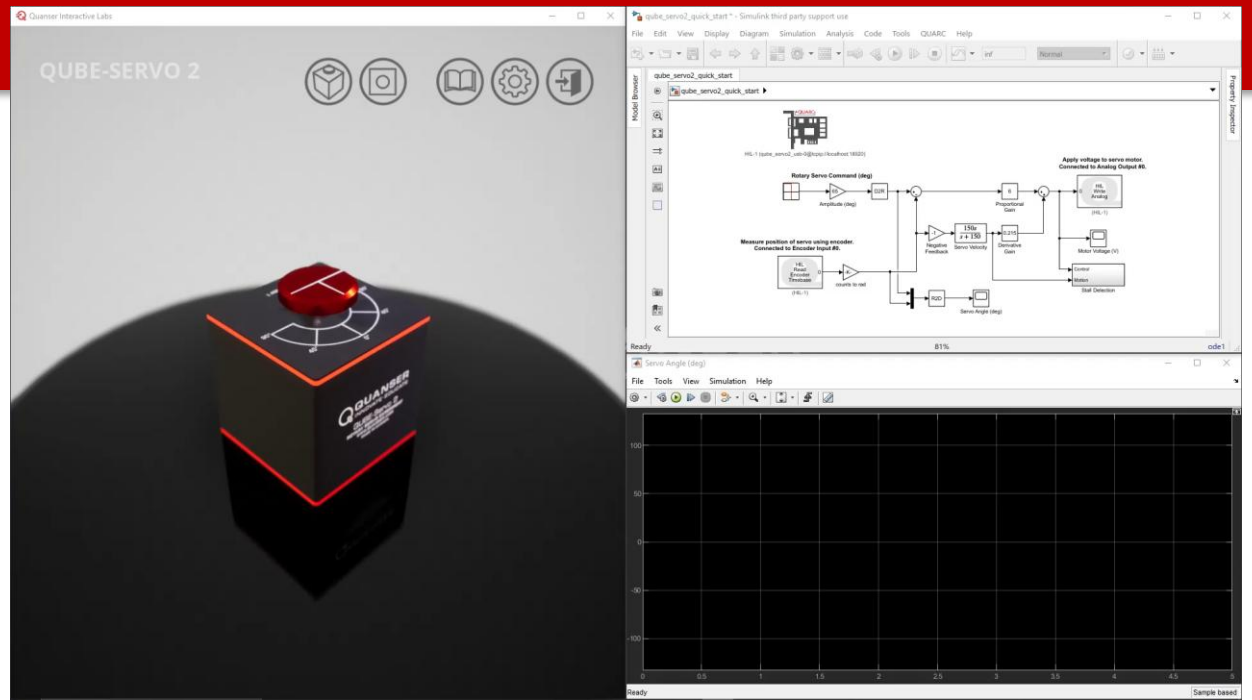
Freescale



Zynq SDR

MathWorks supports project-based learning through **Quanser virtual labs** and **hardware**

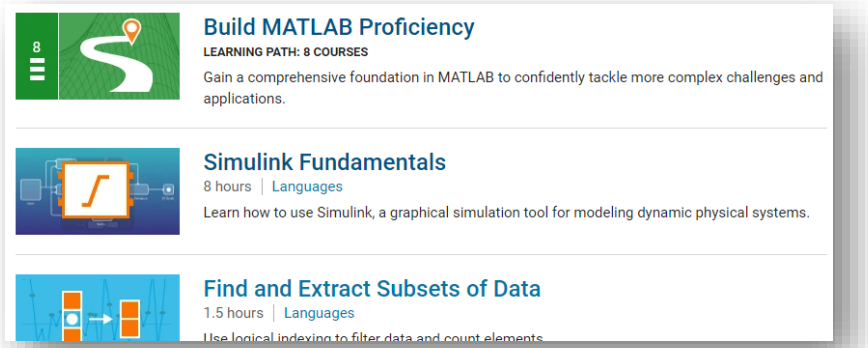
Quanser Interactive Labs is a hardware-inspired, **scalable** way to bring credible, high-fidelity interactive lab experiences to students in **blended, hybrid, or remote laboratory** configurations.





Self-Paced Online Courses

http://matlabacademy.mathworks.com



- MATLAB and Simulink online training courses at 44 Short Courses
  - 12 Learning Paths
- Offers learning flexibility for:
  - upskilling/reskilling
  - exploring new topics and ideas
  - meeting prerequisites for instructor-led courses or consulting engagements
- Choose to take individual lessons, short courses, or full learning paths
- Wide range of available topics—new courses and topics added every few months

Short courses (1-3 hours)

- MATLAB Onramp
- Simulink Onramp
- Circuit Simulation Onramp
- Machine Learning Onramp
- Power Systems Simulation Onramp
- Image Processing Onramp
- System Composer Onramp
- Object Oriented Programming Onramp
- App Building Onramp
- Simscape Onramp
- Stateflow Onramp

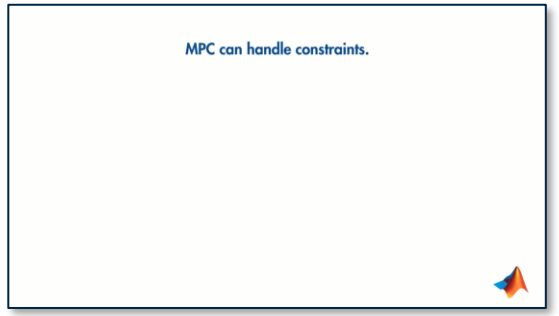
Long courses (7-10 hours)

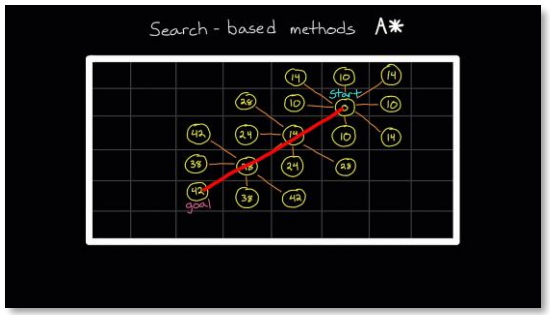
- Simulink Fundamentals
- Image Processing with MATLAB
- Deep Learning with MATLAB
- Machine Learning with MATLAB

Supplement your teaching of engineering concepts

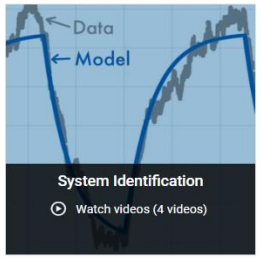
MATLAB Tech Talks: Educational videos for students

- Tech talks help students gain intuition into complex engineering concepts with **easy-to-understand examples**



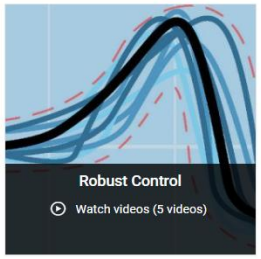


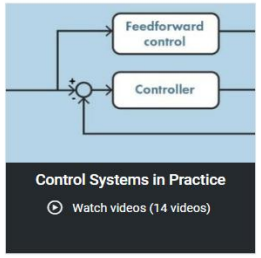
- Visit the MATLAB YouTube channel and Control Systems Tech Talk Library for existing and new videos!

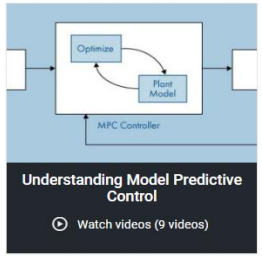


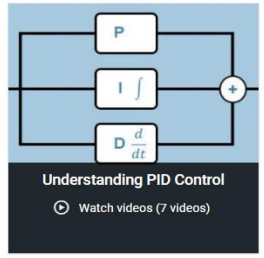




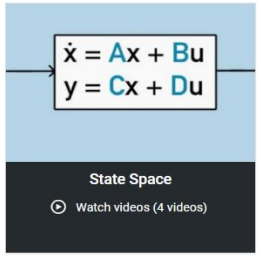


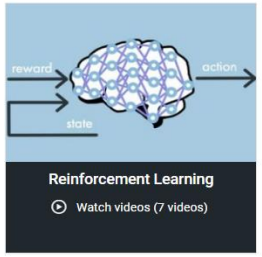


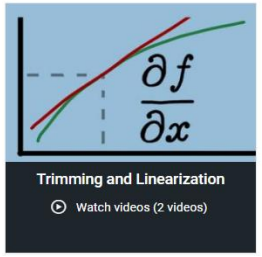




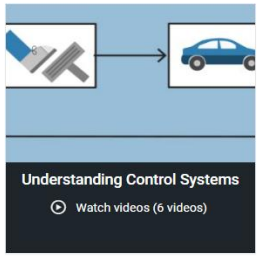


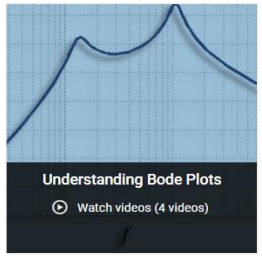


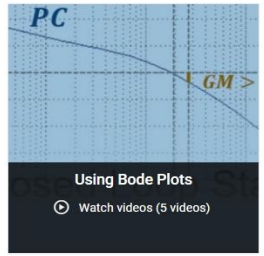






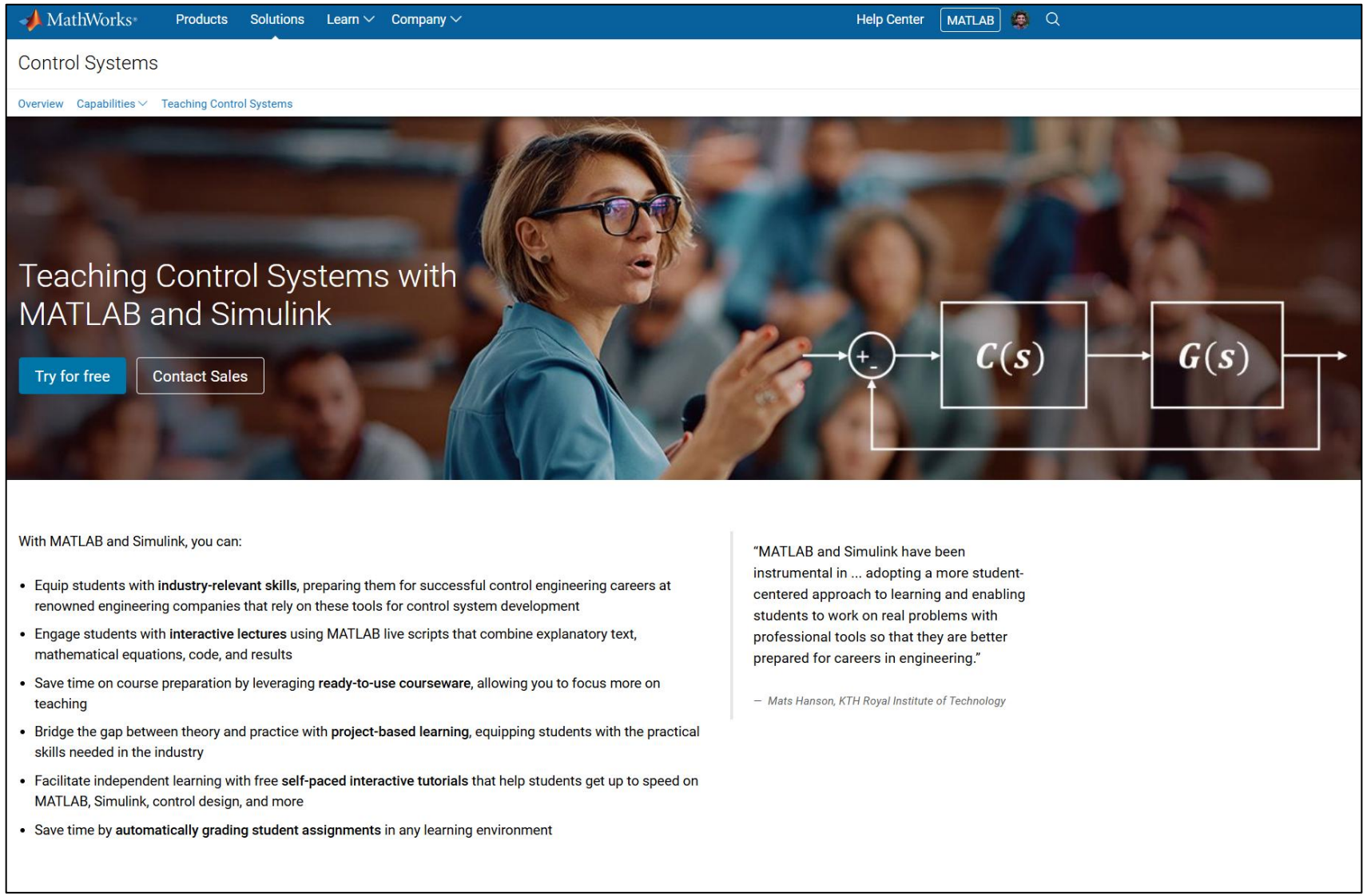








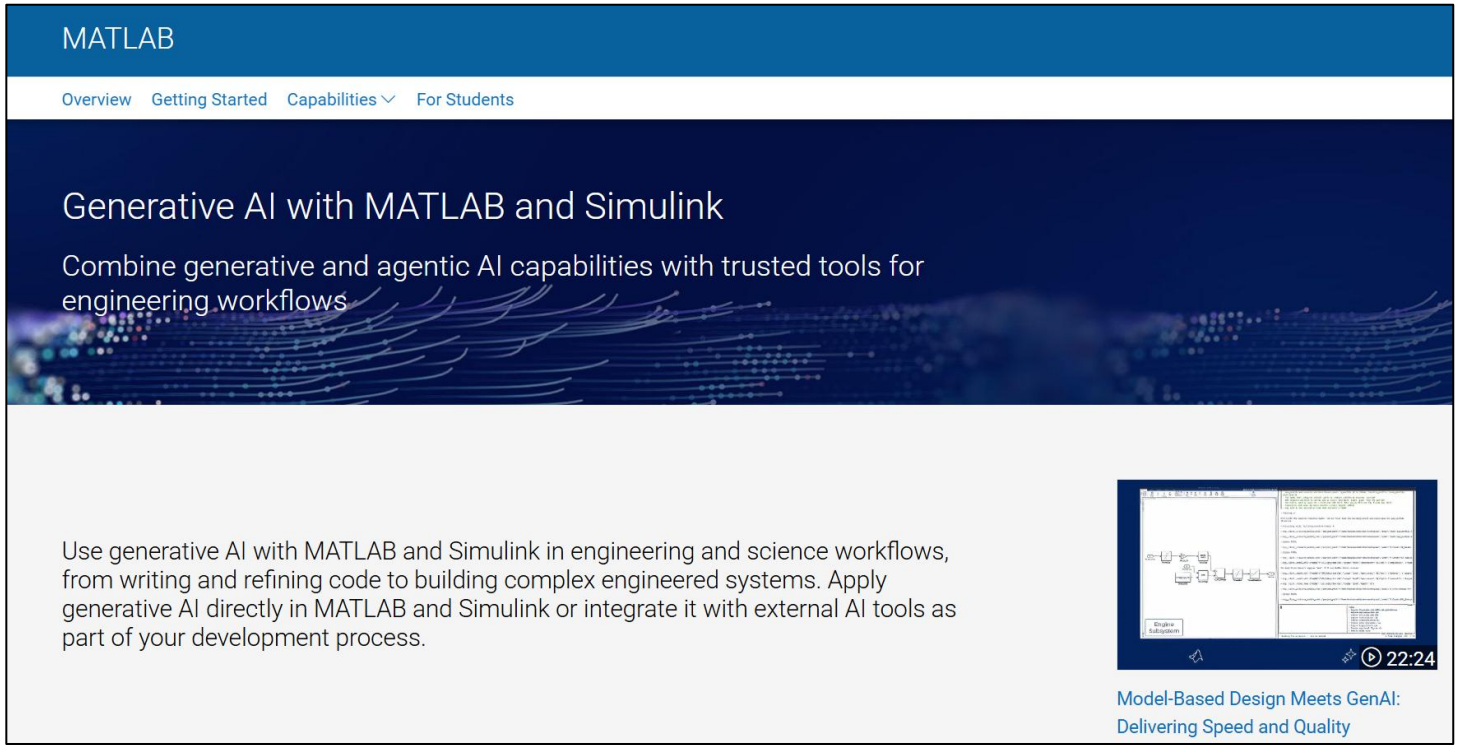
Teaching control systems with MATLAB and Simulink



Link

GenAI and Agentic AI offerings from MathWorks

- GenAI assistants
  - MATLAB Copilot
  - Simulink Copilot
- Agentic AI
  - MATLAB Agentic Toolkit
  - Simulink Agentic Toolkit



MathWorks Education Application Engineer *sjawahar@mathworks.com*

consult with faculty and researchers to support them with their STEM initiatives,

including integrating computational or systems thinking into their curriculum and research









Please take a minute to fill out a survey!



[Survey - From Control Theory to Practice: Addressing Misconceptions in Control Design – Fill out form]